

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: Alibaba B2B E-Commerce Database Management System

Case Study: JBL Bluetooth Earbuds Wholesale Sourcing (Supplier = Seller entity)

1. Database Design

1.1 Supplier (Seller) Table

Attribute	Key Type
Supplier_ID	Primary Key
Supplier_Name	Not Null
Location	Not Null
MOQ_Policy	-
Certification_Status	Check Key
Contact_No	Unique Key
Email	Unique Key

1.2 Inventory Table

Attribute	Key Type
Inventory_ID	Primary Key
Supplier_ID	Foreign Key
Product_ID	Foreign Key
Stock_Quantity	Default, Check Key
Stock_Status	Default, Check Key
Last_Updated	Not Null

2. Primary and Foreign Key Relationship

Supplier → Inventory

- Supplier.Supplier_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Supplier_ID → Foreign Key referencing Supplier.Supplier_ID
- Inventory.Product_ID → Foreign Key referencing Product.Product_ID
- Relationship: One-to-Many (1:M) — one supplier can have multiple inventory records.

3. Table Creation

3.1 Create Supplier Table

```
CREATE TABLE Supplier (  
  Supplier_ID      NUMBER PRIMARY KEY,  
  Supplier_Name    VARCHAR2(100) NOT NULL,  
  Location         VARCHAR2(100) NOT NULL,  
  MOQ_Policy       VARCHAR2(100),  
  Certification_Status VARCHAR2(30)  
    CHECK (Certification_Status IN ('Verified', 'Pending')),  
  Contact_No       VARCHAR2(20) UNIQUE,  
  Email            VARCHAR2(100) UNIQUE  
);
```

Table created.

The Supplier table is created with Supplier_ID as the primary key. It stores supplier details such as supplier name, location, MOQ policy, certification status, contact number, and email. Contact_No and Email are unique, and Certification_Status is restricted by a CHECK constraint.

3.2 Create Inventory Table

```
CREATE TABLE Inventory (  
  Inventory_ID     NUMBER PRIMARY KEY,  
  Supplier_ID      NUMBER NOT NULL,  
  Product_ID       NUMBER NOT NULL,  
  Stock_Quantity   NUMBER DEFAULT 0 CHECK (Stock_Quantity >= 0),  
  Stock_Status     VARCHAR2(20) DEFAULT 'In Stock'  
    CHECK (Stock_Status IN ('In Stock', 'Out of Stock')),  
  Last_Updated     DATE NOT NULL,  
  CONSTRAINT fk_inv_supplier FOREIGN KEY (Supplier_ID)  
    REFERENCES Supplier(Supplier_ID),  
  CONSTRAINT fk_inv_product FOREIGN KEY (Product_ID)  
    REFERENCES Product(Product_ID)  
);
```

Table created.

The Inventory table is created with Inventory_ID as the primary key. Supplier_ID and Product_ID are foreign keys that establish relationships with the Supplier and the existing Product table. Stock_Quantity defaults to 0, Stock_Status defaults to 'In Stock', and CHECK constraints prevent negative stock and invalid status values.

4. Data Insertion

4.1 Supplier Records

Five supplier records are inserted into the Supplier table, followed by a COMMIT to save the transaction.

```
INSERT INTO Supplier
VALUES (1, 'Shenzhen SoundWave Electronics', 'Shenzhen, China',
       'MOQ 500 pcs', 'Verified', '+8613800138001',
       'sales@soundwave-sz.com');
1 row created.

INSERT INTO Supplier
VALUES (2, 'Guangzhou AudioTech Co.', 'Guangzhou, China',
       'MOQ 300 pcs', 'Verified', '+8613800138002',
       'info@audiotech-gz.com');
1 row created.

INSERT INTO Supplier
VALUES (3, 'Dongguan BassPro Trading', 'Dongguan, China',
       'MOQ 1000 pcs', 'Pending', '+8613800138003',
       'contact@basspro-dg.com');
1 row created.

INSERT INTO Supplier
VALUES (4, 'Huizhou EarPod Manufacturing', 'Huizhou, China',
       'MOQ 200 pcs', 'Verified', '+8613800138004',
       'export@earpod-hz.com');
1 row created.

INSERT INTO Supplier
VALUES (5, 'Foshan Wireless Audio Ltd.', 'Foshan, China',
       'MOQ 500 pcs', 'Verified', '+8613800138005',
       'hello@wirelessaudio-fs.com');
1 row created.

COMMIT;
Commit complete.
```

4.2 Inventory Records

Inventory records are inserted with supplier IDs, product IDs, stock quantities, stock status, and last updated dates. A COMMIT saves the transaction.

```
INSERT INTO Inventory
VALUES (101, 1, 101, 5000, 'In Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (102, 1, 102, 0, 'Out of Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (103, 2, 103, 3000, 'In Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (104, 2, 104, 1500, 'In Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (105, 3, 105, 0, 'Out of Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (106, 3, 106, 2500, 'In Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (107, 4, 107, 4000, 'In Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (108, 4, 108, 0, 'Out of Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (109, 5, 109, 2000, 'In Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (110, 5, 110, 0, 'Out of Stock',
       TO_DATE('19-09-2026','DD-MM-YYYY'));
1 row created.

COMMIT;
Commit complete.
```

5. Data Retrieval

5.1 Display Supplier Records

All supplier records are retrieved to verify successful insertion.

```
Select * from Supplier;
```

SUPPLIER_ID	SUPPLIER_NAME	LOCATION	MOQ_POLICY	CERTIFICATION_STATUS	CONTACT_NO	EMAIL
1	Shenzhen SoundWave Electronics	Shenzhen, China	MOQ 500 pcs	Verified	+8613800138001	sales@soundwave-sz.com
2	Guangzhou AudioTech Co.	Guangzhou, China	MOQ 300 pcs	Verified	+8613800138002	info@audiotech-gz.com
3	Dongguan BassPro Trading	Dongguan, China	MOQ 1000 pcs	Pending	+8613800138003	contact@basspro-dg.com
4	Huizhou EarPod Manufacturing	Huizhou, China	MOQ 200 pcs	Verified	+8613800138004	export@earpod-hz.com
5	Foshan Wireless Audio Ltd.	Foshan, China	MOQ 500 pcs	Verified	+8613800138005	hello@wirelessaudio-fs.com

5 rows selected.

5.2 Display Inventory Records

All inventory records are retrieved to verify the stock information.

```
Select * from Inventory;
```

INVENTORY_ID	SUPPLIER_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDATED
101	1	101	5000	In Stock	19-SEP-26
102	1	102	0	Out of Stock	19-SEP-26
103	2	103	3000	In Stock	19-SEP-26
104	2	104	1500	In Stock	19-SEP-26
105	3	105	0	Out of Stock	19-SEP-26
106	3	106	2500	In Stock	19-SEP-26
107	4	107	4000	In Stock	19-SEP-26
108	4	108	0	Out of Stock	19-SEP-26
109	5	109	2000	In Stock	19-SEP-26
110	5	110	0	Out of Stock	19-SEP-26

10 rows selected.

6. Supplier Product Information

A JOIN query is used to display the supplier ID, supplier name, product name, stock quantity, stock status, and last updated date.

```
Select
s.Supplier_ID,
s.Supplier_Name,
p.Product_Name,
i.Stock_Quantity,
i.Stock_Status,
i.Last_Updated
From Supplier s
join Inventory i
ON s.Supplier_ID = i.Supplier_ID
join Product p
ON i.Product_ID = p.Product_ID
Order by s.Supplier_Name, i.Inventory_ID;
```

SUPPLIER_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDATED
3	Dongguan BassPro Trading	JBL Tune Flex	0	Out of Stock	19-SEP-26
3	Dongguan BassPro Trading	JBL Vibe Beam	2500	In Stock	19-SEP-26
5	Foshan Wireless Audio Ltd.	JBL Wave Buds 2	2000	In Stock	19-SEP-26
5	Foshan Wireless Audio Ltd.	JBL Tune Buds	0	Out of Stock	19-SEP-26
2	Guangzhou AudioTech Co.	JBL Tune 230NC TWS	3000	In Stock	19-SEP-26
2	Guangzhou AudioTech Co.	JBL Wave Beam	1500	In Stock	19-SEP-26
4	Huizhou EarPod Manufacturing	JBL Endurance Peak 3	4000	In Stock	19-SEP-26
4	Huizhou EarPod Manufacturing	JBL Live Buds 3	0	Out of Stock	19-SEP-26
1	Shenzhen SoundWave Electronics	JBL Tune 130NC TWS	5000	In Stock	19-SEP-26
1	Shenzhen SoundWave Electronics	JBL Wave 200TWS	0	Out of Stock	19-SEP-26

10 rows selected.

7. Product Availability

7.1 Available Products

Products with stock available are displayed using the Stock_Status field.

```
Select
p.Product_ID,
p.Product_Name,
i.Stock_Quantity,
i.Stock_Status
From Product p
join Inventory i
ON p.Product_ID = i.Product_ID
where i.Stock_Status = 'In Stock';
```

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
101	JBL Tune 130NC TWS	5000	In Stock
103	JBL Tune 230NC TWS	3000	In Stock
104	JBL Wave Beam	1500	In Stock
106	JBL Vibe Beam	2500	In Stock
107	JBL Endurance Peak 3	4000	In Stock
109	JBL Wave Buds 2	2000	In Stock

6 rows selected.

7.2 Unavailable Products

Products with zero stock or Out of Stock status are displayed separately.

```
Select
p.Product_ID,
p.Product_Name,
i.Stock_Quantity,
i.Stock_Status
From Product p
join Inventory i
ON p.Product_ID = i.Product_ID
where i.Stock_Status = 'Out of Stock';
```

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
102	JBL Wave 200TWS	0	Out of Stock
105	JBL Tune Flex	0	Out of Stock
108	JBL Live Buds 3	0	Out of Stock
110	JBL Tune Buds	0	Out of Stock

4 rows selected.

8. Inventory Update

Demonstrate updating stock information: two out-of-stock products (Inventory_ID 110 and 102) are restocked, and their status and last updated date are refreshed. The update is verified with a SELECT query.

```
UPDATE Inventory
SET Stock_Quantity = 3000,
    Stock_Status = 'In Stock',
    Last_Updated = TO_DATE('20-09-2026', 'DD-MM-YYYY')
WHERE Inventory_ID = 110;
```

1 row updated.

```
UPDATE Inventory
SET Stock_Quantity = 1000,
    Stock_Status = 'In Stock',
    Last_Updated = TO_DATE('20-09-2026', 'DD-MM-YYYY')
WHERE Inventory_ID = 102;
```

1 row updated.

```
COMMIT;
Commit complete.
```

```
SELECT *
FROM Inventory
WHERE Inventory_ID IN (102, 110);
```

INVENTORY_ID	SUPPLIER_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDATED
102	1	102	1000	In Stock	20-SEP-26
110	5	110	3000	In Stock	20-SEP-26

2 rows selected.

9. Inventory Status Reports

9.1 Complete Inventory Report

Combines the Inventory, Supplier and Product tables to show the current stock position of every item (after the update in Section 8).

```
SELECT
    i.Inventory_ID,
    s.Supplier_Name,
    p.Product_Name,
    i.Stock_Quantity,
    i.Stock_Status,
    i.Last_Updated
FROM Inventory i
JOIN Supplier s
ON i.Supplier_ID = s.Supplier_ID
JOIN Product p
ON i.Product_ID = p.Product_ID
ORDER BY i.Inventory_ID;
```

INVENTORY_ID	SUPPLIER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDATED
101	Shenzhen SoundWave Electronics	JBL Tune 130NC TWS	5000	In Stock	19-SEP-26
102	Shenzhen SoundWave Electronics	JBL Wave 200TWS	1000	In Stock	20-SEP-26
103	Guangzhou AudioTech Co.	JBL Tune 230NC TWS	3000	In Stock	19-SEP-26
104	Guangzhou AudioTech Co.	JBL Wave Beam	1500	In Stock	19-SEP-26
105	Dongguan BassPro Trading	JBL Tune Flex	0	Out of Stock	19-SEP-26
106	Dongguan BassPro Trading	JBL Vibe Beam	2500	In Stock	19-SEP-26
107	Huizhou EarPod Manufacturing	JBL Endurance Peak 3	4000	In Stock	19-SEP-26
108	Huizhou EarPod Manufacturing	JBL Live Buds 3	0	Out of Stock	19-SEP-26
109	Foshan Wireless Audio Ltd.	JBL Wave Buds 2	2000	In Stock	19-SEP-26
110	Foshan Wireless Audio Ltd.	JBL Tune Buds	3000	In Stock	20-SEP-26

10 rows selected.

9.2 Total Stock by Supplier

The total stock quantity held by each supplier is calculated using SUM with GROUP BY.

```
Select
s.Supplier_Name,
Sum(i.Stock_Quantity) as Total_Stock
From Supplier s
join Inventory i
ON s.Supplier_ID = i.Supplier_ID
Group by s.Supplier_Name
Order by s.Supplier_Name;
```

SUPPLIER_NAME	TOTAL_STOCK
Dongguan BassPro Trading	2500
Foshan Wireless Audio Ltd.	5000
Guangzhou AudioTech Co.	4500
Huizhou EarPod Manufacturing	4000
Shenzhen SoundWave Electronics	6000

5 rows selected.

9.3 Product with Zero Stock

Products whose stock quantity is zero are listed along with the supplier responsible for restocking them.

```
SELECT
    p.Product_ID,
    p.Product_Name,
    s.Supplier_Name,
    i.Stock_Quantity,
    i.Stock_Status
FROM Product p
JOIN Inventory i
ON p.Product_ID = i.Product_ID
JOIN Supplier s
ON i.Supplier_ID = s.Supplier_ID
WHERE i.Stock_Quantity = 0;
```

PRODUCT_ID	PRODUCT_NAME	SUPPLIER_NAME	STOCK_QUANTITY	STOCK_STATUS
105	JBL Tune Flex	Dongguan BassPro Trading	0	Out of Stock
108	JBL Live Buds 3	Huizhou EarPod Manufacturing	0	Out of Stock

2 rows selected.

9.4 Available and Unavailable Product Count

The number of inventory records in each stock status is counted using GROUP BY.

```
Select Stock_Status, Count(*) as Total_Product
From Inventory
Group By Stock_Status
Order By Stock_Status;
```

STOCK_STATUS	TOTAL_PRODUCT
In Stock	8
Out of Stock	2

2 rows selected.

9.5 Low Stock Report

Products that are still available but have fewer than 2,000 units left are flagged so that buyers and suppliers can plan restocking.

```
SELECT
    p.Product_ID,
    p.Product_Name,
    s.Supplier_Name,
    i.Stock_Quantity
FROM Product p
JOIN Inventory i
ON p.Product_ID = i.Product_ID
JOIN Supplier s
ON i.Supplier_ID = s.Supplier_ID
WHERE i.Stock_Quantity > 0
AND i.Stock_Quantity < 2000
ORDER BY i.Stock_Quantity;
```

PRODUCT_ID	PRODUCT_NAME	SUPPLIER_NAME	STOCK_QUANTITY
102	JBL Wave 200TWS	Shenzhen SoundWave Electronics	1000
104	JBL Wave Beam	Guangzhou AudioTech Co.	1500

2 rows selected.

10. Conclusion

The Supplier and Inventory tables were successfully created and connected with the existing Product table using primary and foreign keys. Supplier and product information can be maintained along with stock quantities and stock status. Available and unavailable products can be tracked using SQL queries. Inventory reports such as complete inventory, supplier-wise stock, availability, and low-stock reports can also be generated for the Alibaba B2B wholesale sourcing platform.